University Homework & Quiz LaTeX Template

A General-Purpose Bilingual Academic Template for Assignments and Quizzes

University of Tehran

1 Introduction

This repository provides a versatile, general-purpose **academic document template** designed for university courses across computer science, engineering, and mathematics. Originally adapted for courses at the **University of Tehran**, it is fully course-agnostic and provides two specialized document classes:

- `homework.cls` — For comprehensive homework assignments, problem sets, and lab handouts with multiple official header styles.
- `quiz.cls` — For short exams, quizzes, and classroom assessments with student information fields and framed answer boxes.

2 Key Features

- **Dual Document Classes:** Specialized classes for both homework assignments and quizzes.
- **Multiple Header Styles (`homework.cls`):** Switch between Classic Banner (`header-type=1`) and Modern Wide Side-by-Side (`header-type=2`) with large 2.0cm logos.
- **Quiz Environment (`quiz.cls`):** Dedicated header with exam duration (`\quiztime`), student name, student ID fields, and answer boxes (`\answerbox`).
- **Smart Asset & Font Locator:** Automatically searches up to 5 nested folder levels for `Fonts/` and `Background/` assets.
- **Bilingual & Persian Typesetting (XePersian):** Bundled standard Persian fonts (*XB Niloofar*, *XB Zar*, *IRZar*, *IRLotus*, *XB Yas*).
- **Modular Subfiles Structure:** Organize questions into standalone subfiles combined seamlessly via `\subfile{...}`.
- **Conditional Solutions:** Toggle student solutions globally using the `compileanswers` boolean flag.
- **Syntax-Highlighted Code:** Formats source code elegantly using `minted` for Python, C/C++, Verilog, Java, etc.

3 Repository Structure

- `homework.cls` — Homework document class supporting multiple header designs and margin extensions.
- `quiz.cls` — Quiz document class supporting student information fields and answer boxes.
- `Background/` — High-resolution University of Tehran seal and Faculty of Engineering emblem.
- `Fonts/` — High-quality Persian TrueType fonts bundled directly in the repository.
- `HW.tex` — Master document demonstrating homework layout with `header-type=2`.
- `latexmkrc` — Automated build configuration for Overleaf and local `latexmk` XeLaTeX builds.
- `LaTeX-build/` — Standard build directory for compiler artifacts (`.pdf`, `.log`, `.aux`).

4 Document Class Options

- `largemargins` — Widens page margins to leave ample space for grading annotations (both classes).
- `newpage` — Automatically begins each new question on a fresh page (both classes).
- `anonymous` — Omits personal contributor details for blind grading (both classes).
- `header-type=1|2` — Selects Classic Banner (`1`) or Modern Wide (`2`) header in `homework.cls`.
- `nostudentinfo` — Suppresses the student name, ID, and duration fields in `quiz.cls`.

5 Homework Configuration Example (`homework.cls`)

```
\documentclass[12pt,largemargins,header-type=2]{homework}

\newcommand{\eventclass}{Automata & Language Theory} % Course name
\newcommand{\eventsemester}{Spring 2026}             % Semester
\newcommand{\eventtype}{Homework}                    % Assignment type
\newcommand{\eventnum}{2}                            % Assignment number
\newcommand{\eventinstructor}{Dr. Fadaei & Dr. Yaghoobzadeh} % Instructors
\newcommand{\eventcntrbs}{Mohammad Khoshroo}         % Author or TA
\newcommand{\hwemail}{ta-email@ut.ac.ir}             % Contact email
\newcommand{\eventdatetype}{Due Date:}
\newcommand{\eventDeadlineDay}{8}
\newcommand{\eventDeadlineMonth}{May}
\newcommand{\eventDeadlineYear}{2026}
\newcommand{\eventDeadlineHour}{23:59}

\begin{document}
\makeheader
\subfile{Q-A/Q1.tex}
\clearpage
\subfile{Q-A/Q2.tex}
\end{document}
```

6 Quiz Configuration Example (`quiz.cls`)

```
\documentclass[12pt,largemargins]{quiz}

\newcommand{\eventclass}{Discrete Mathematics}      % Course name
\newcommand{\eventtype}{Quiz}                       % Exam type
\newcommand{\eventnum}{9}                           % Quiz number
\newcommand{\eventname}{Number Theory}              % Topic
\newcommand{\eventcntrbs}{Instructor: F. Mohammadi} % Instructor
\newcommand{\eventdatetype}{Date:}                  % Date prefix
\newcommand{\eventdate}{2026/05/10}                 % Date
\newcommand{\quiztime}{15 Minutes}                  % Exam duration

\begin{document}
\maketitle

\question
Find all positive integers $a$ and $b$ such that:
\[ \frac{1}{a} + \frac{1}{b} = \frac{1}{101} \]

\answerbox{2.5cm} % Framed answer box for students

\solution
Detailed solution is written here.
\end{document}
```

7 Metadata Variables Reference

- `\eventclass{...}` — Course title (e.g. Automata Theory, AI).
- `\eventtype{...}` — Assignment or quiz type (e.g. Homework, Quiz, Project).
- `\eventnum{...}` — Assignment or quiz number.
- `\eventname{...}` — Topic or chapter subtitle.
- `\eventsemester{...}` — Academic semester (e.g. Spring 2026).
- `\eventinstructor{...}` — Instructor name(s).
- `\eventinstructortype{...}` — Custom instructor label prefix (default: Instructor).
- `\eventcntrbs{...}` — Teaching assistants, designers, or authors.

- `\eventcntrbstype{...}` — Custom contributor label prefix (default: Designers).
- `\hwemail{...}` — Contact email address (automatically wrapped in a `mailto:` link).
- `\eventdatetype{...}` — Date label prefix (default: Due Date: or Date:).
- `\eventdate{...}` — Unified date string.
- `\eventDeadlineDay`, `\eventDeadlineMonth`, `\eventDeadlineYear`, `\eventDeadlineHour` — Granular deadline components.
- `\eventcourselink{...}` — URL pointing to the course web page.
- `\quiztime{...}` — Quiz duration / allowed response time (e.g. 15 Minutes).
- `\quiztimelabel`, `\studentnamelabel`, `\studentidlabel` — Customizable field labels for quizzes.

8 Question, Solution & Utility Commands

- `\makeheader` or `\maketitle` — Generates the official course header box (both classes).
- `\setheadertype{1|2}` — Dynamically switches header style at runtime (`homework.cls`).
- `\question` — Starts a numbered question with a colored dividing rule.
- `\question*{Title}` — Starts a question header with a custom title string.
- `\question#{Points}` — Creates a question header with point allocation badge (`homework.cls`).
- `\solution` — Creates a solution header with a red dividing rule.
- `\solution*{Title}` — Custom solution header.
- `\answerbox{height}` — Generates an empty framed box of specified height for student answers (both classes).
- `\ExternalLink` — Renders an inline external link TikZ icon.
- `\RN{number}` — Converts an integer to uppercase Roman numerals (e.g. `\RN{4}` gives IV).
- `alphaparts` / `arabicparts` — Sub-question environments for (a), (b) or 1.1, 1.2 numbering.
- `\questiontype` / `\solutiontype` — Prefix redefinitions for questions and solutions.
- `\writtensection` — Section numbering prefix (e.g. `5` for 5.1, 5.2).
- `\ifcompileanswers` — Boolean conditional for toggling solution visibility.

9 Code Listings with Minted

To include syntax-highlighted code in assignments or solutions, use the `minted` environment (wrap within `\LRE{}` when inside a Persian context):

```
import numpy as np

def solve_system(A, b):
    x = np.linalg.solve(A, b)
    residual = np.linalg.norm(np.dot(A, x) - b)
    return x, residual
```

10 Subfiles & Build Setup

The template supports modular multi-file assignments with the `subfiles` package:

- Always set `%!TEX root = ../HW.tex` at the top of question files in `Q-A/`.

- In VS Code LaTeX Workshop, keep `"latex-workshop.latex.rootFile.useSubFile": false` so the master document is always built.
- Use `latexmkrc` for automatic XeLaTeX configuration on Overleaf.